small to be resolved by a telescope, but the star's total brightness can be increased by a factor of 10 or 100 or more.

If the star and the hole are far apart as in Figure 8.2, however, the strength of the focusing can be large, an increase of 10 or 100 or more in stellar brightness. But interstellar distances are so vast that the necessary Earth–hole–star lineup would be an exceedingly rare event, so rare that to search for one would be hopeless. Moreover, even if such a lensing were observed, the light rays from star to Earth would pass the hole at so large a distance (Figure 8.2) that there would be room for an entire star to sit at the hole's location and act as the lens. An astronomer on Earth thus could not know whether the lens was a black hole or merely an ordinary, but dim, star.

Zel'dovich must have gone through a chain of reasoning much like this as he sought a method to observe black holes. His chain led finally to a method with some promise (Figure 8.3): Suppose that a black hole and a star are in orbit around each other (they form a *binary system).* When astronomers train their telescopes on this binary, they will see light from only the star; the hole will be invisible. However, the star's light will give evidence of the hole's presence: As the star moves around the hole in its orbit, it will travel first toward the Earth and then away. When it is traveling toward us, the Doppler effect should shift the star's light toward the blue, and when moving away, toward the red. Astronomers can measure such shifts with high precision, since the star's light, when sent through a spectrograph (a sophisticated form of prism), exhibits sharp spectral lines, and a slight change in the wavelength (color) of such a line stands out clearly. From a measurement of the shift in wavelength, astronomers can infer the velocity of the star toward or away from Earth, and by monitoring the shift as time passes, they can infer how the star's velocity changes with time. The magnitude of those changes might typically be somewhere between 10 and 100 kilometers per second, and the accuracy of the measurements is typically 0.1 kilometer per second.

What does one learn from such high-precision measurements of the star's velocity? One learns something about the mass of the hole: The more massive is the hole, the stronger is its gravitational pull on the star, and thus the stronger must be the centrifugal forces by which the star resists getting pulled into the hole. To acquire strong centrifugal forces, the star must move rapidly in its orbit. Thus, large orbital velocity goes hand in hand with large black-hole mass.